Satellite-Guided Bias Correction and Probabilistic Forecasting of CMIP6 Sea Level Anomaly Using Tube Loss Based Neural Networks

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***Abstract*—Accurate projections of sea level anomaly (SLA) are critical for coastal risk assessment under climate change. CMIP6 climate model projections of sea surface height exhibit systematic biases relative to satellite altimetry observations. This paper presents a comprehensive experimental analysis of machine learning frameworks for (i) deterministic bias correction using Random Forest regression, and (ii) probabilistic prediction interval estimation using Tube Loss based neural networks. Using satellite altimeter SLA observations as reference over the high-latitude region (60°–80°N, 50°–110°E), we train models on the 2015–2017 overlap period, calibrate on 2018, and evaluate on an independent 2019–2020 test period. The Random Forest correction reduces RMSE by 44.2% (from 0.1037m to 0.0578m). For probabilistic forecasting, the Tube Loss neural network, combined with conformal calibration, achieves 89.3% coverage for 90% prediction intervals and 92.3% for 95% intervals. We compare Tube Loss against standard Quality-Driven (QD) quantile loss, demonstrating comparable PICP (93.0%) and MPIW (0.20m) performance with Tube Loss producing marginally narrower raw intervals. Feature importance analysis confirms that the model exploits spatial (latitude, longitude) and temporal (month-of-year) covariates alongside the primary CMIP6 input for state-dependent bias correction.**

***Index Terms*—sea level anomaly, CMIP6, bias correction, satellite altimetry, Tube Loss, prediction intervals, quantile regression, Random Forest, neural networks**

I. Introduction

Sea level rise is among the most consequential impacts of anthropogenic climate change, with direct implications for coastal flooding, erosion, and infrastructure planning [1]. Coupled Model Intercomparison Project Phase 6 (CMIP6) Earth System Models provide projections of sea surface height (SSH) under various Shared Socioeconomic Pathways (SSPs), but these projections exhibit systematic biases relative to satellite observations [2].

Classical bias correction methods (delta method, empirical quantile mapping) are univariate and assume stationary bias structures [3]. Recent machine learning approaches have demonstrated superior performance by exploiting spatial, temporal, and state-dependent bias patterns [4]. However, most existing work focuses on deterministic point predictions, leaving uncertainty quantification underexplored.

This paper makes two contributions: (1) a Random Forest deterministic bias correction framework for CMIP6 SLA, and (2) a Tube Loss [5] based neural network for probabilistic prediction interval estimation, benchmarked against standard quantile decomposition (QD) loss. The Tube Loss function, recently proposed for prediction interval estimation, offers theoretical coverage guarantees and allows trading off interval width versus calibration through a single optimization [5].

II. Related Work

Bias correction of climate model projections and uncertainty quantification for sea level have been active research areas, spanning classical statistical methods, modern machine learning approaches, and probabilistic forecasting frameworks. We organize the literature into five categories and identify the specific gaps that our work addresses.

A. Classical Statistical Bias Correction

The simplest bias correction method is the **delta method**, which applies a constant additive or multiplicative shift derived from the historical period. While widely used for temperature and precipitation [2], this method assumes stationary bias and cannot capture non-linear, state-dependent, or spatially varying errors [3]. Lemos et al. [2] applied a modified quantile mapping to CMIP5 projected wave climate, achieving 10–20% RMSE reductions for significant wave height, but noted that the delta method's stationarity assumption degrades under climate change.

Empirical Quantile Mapping (EQM) corrects the full distribution by mapping model quantiles to observed quantiles. A comprehensive intercomparison of EQM, Detrended Quantile Mapping (DQM), and Quantile Delta Mapping (QDM) across 11 CMIP6 GCMs on six continents found that EQM achieved the best overall RMSE of 0.30 for precipitation with the lowest uncertainty (variance = 0.0027) [11]. **QDM** [3] preserves relative trends better in projection periods, but a recent "perfect model" experiment [12] demonstrated that standard quantile mapping introduces biases in means, standard deviations, and extremes when applied to future climate scenarios, failing to capture the climate change signal. All these methods are univariate, treat each grid point independently, and cannot exploit spatial or temporal covariates.

For sea level specifically, bias correction has received less attention than for temperature and precipitation. The Ensemble Time-series Quantile Mapping (ETQM) approach [22] combines multiple methods to correct CMIP6 outputs over the Indian Ocean, reducing extremes biases by 1.5–4.5% for temperature. However, no comparable quantile mapping study has targeted sea level anomaly with the RMSE reductions we achieve.

B. Machine Learning for Ocean and Sea Level Bias Correction

Machine learning methods have demonstrated clear advantages over classical approaches for ocean variable correction. We review the key contributions organized by target variable and methodology.

1) Sea Level Applications: Doganay et al. [13] developed a state-dependent ML bias correction for sea level applied to an ensemble of downscaled regional climate projections over the Swedish Baltic coast, producing 2,600+ model-years of corrected hourly sea levels. Their method conditions corrections on the model's internal state rather than output distributions, improving upon classical approaches on standard skill scores. However, this method provides only deterministic corrections without uncertainty quantification.

Pasula and Subramani [23] applied deep learning (UNet, BiLSTM, ConvLSTM) to correct CMIP6 projections of sea surface temperature (SST) and dynamic sea level (DSL) in the Bay of Bengal. Starting from raw CMIP6 errors of 1.2°C (SST) and 1.1m (DSL) against ORAS5 reanalysis, their deep learning approach reduced RMSE by 0.15°C for SST and 0.3m for DSL compared to the Equi-Distant CDF (EDCDF) statistical baseline, representing approximately 15% improvement. Our method achieves a substantially higher 44.2% RMSE reduction, though we note the domains and variables differ.

2) Sea Surface Height Reconstruction: Martin et al. [24] used deep learning to synthesize satellite altimetry and SST observations for SSH anomaly mapping. Their approach achieved 17% lower RMSE than optimal interpolation in the Gulf Stream Extension region and resolved spatial scales 30% smaller than conventional methods. This work targets SSH reconstruction from observations rather than climate model bias correction, but demonstrates the potential of neural networks for improving SSH products.

3) Ocean General Circulation Model Correction: Guillou et al. [25] embedded neural networks directly into the NEMO v4 ocean model to correct sea surface biases by learning state-dependent errors in air-sea heat fluxes. The pre-trained network uses atmospheric forcing, ocean state, and stratification indices as predictors, with wind forcing emerging as the most important variable. The corrections showed positive impacts beyond the training period for both hindcasts and predictions, demonstrating transferability across time periods.

4) Atmospheric and SWH Applications: Chapman et al. [14] used CNN-based state-dependent corrections to improve atmospheric model biases by 25–35% for land precipitation. Sun et al. [6] achieved 40% RMSE reduction for SWH forecasts over the Northwest Pacific using deep convolutional BU-Net architectures, but required full spatial fields as input rather than the minimal 4-feature set we employ. Ellenson et al. [26] used bagged regression trees to correct WaveWatch III SWH forecasts, reducing RMSE from 0.58m to 0.48m (17% improvement) along the US West Coast, with corrections transferable across buoy locations. Alfaris et al. [27] compared XGBoost and Random Forest for ocean current temperature prediction, while XGBoost has been applied to correct Indian Ocean surface pCO_2 simulations with approximately 40% RMSE improvement [28]. These studies collectively demonstrate that tree-based ensembles are effective for ocean variable correction with minimal feature engineering.

5) Multi-decadal Sea Level Prediction: Hammoud et al. [15] applied neural networks to multi-decadal sea level prediction using CESM2 large ensembles and satellite altimeter data, achieving $R^2 > 0.9$ at 2° resolution with 30-year lead times. Their approach provides uncertainty estimates via spectral clustering but does not produce calibrated prediction intervals.

C. Uncertainty Quantification for Sea Level and Ocean Forecasting

The need for probabilistic predictions with calibrated uncertainty has been increasingly recognized in the ocean and climate communities.

1) Deep Ensemble Methods: Chen et al. [16] developed an LSTM Deep Ensemble (LSTM-DE) for real-time significant wave height prediction, achieving $R^2 > 0.9$ and demonstrating that uncertainty calibration improved prediction interval quality by over 50%. Li et al. [29] applied CNN-BiLSTM hybrid models with modal decomposition for forecasting SWH intervals along China's coast. These approaches use computationally expensive ensemble training, whereas our method produces prediction intervals from a single lightweight network.

2) Emulators and Flow-based Models: Gregory et al. [17] developed ISEFlow, a flow-based neural network emulator for sea level projections that separates data coverage uncertainty from emulator uncertainty using normalizing flows and deep ensemble LSTMs. Feldmann et al. [18] used Gaussian process emulators with uncertainty quantification for Antarctic ice-sheet-driven sea level change. Both approaches target long-term projections rather than monthly bias correction.

3) Conformal Prediction for Weather and Climate: Conformal prediction has recently emerged as a principled framework for uncertainty quantification in weather and climate. Van Straaten et al. [20] applied conformal prediction as a post-processing method for neural weather models, providing calibrated error bounds across all variables, lead times, and spatial locations with negligible computational cost. Bassetti et al. [19] developed conformal ensembles for GCM projections, outperforming inter-model variability methods. Rios et al. [30] proposed meteorologically-informed adaptive conformal prediction for tropical cyclone intensity forecasting, dynamically adjusting uncertainty by incorporating process information such as intensity and evolutionary stage. Our approach integrates conformal calibration with Tube Loss, combining structured interval estimation with distribution-free coverage guarantees.

D. Tube Loss and Prediction Interval Methods

The Tube Loss function [5] was recently proposed as an alternative to the classical pinball (quantile) loss for prediction interval estimation. Unlike traditional quantile regression, which trains separate models for upper and lower quantiles, Tube Loss estimates both bounds simultaneously via a single optimization problem. The loss function includes three key innovations:

1) **Asymmetric inside/outside penalties:** Points outside the interval incur a penalty proportional to q, while points inside incur a penalty proportional to $(1 - q)$, naturally pushing toward target coverage.
2) **Positioning parameter r:** Allows shifting the interval along the response distribution's support to capture denser regions, producing tighter intervals for skewed distributions.
3) **Width penalty δ:** Controls the trade-off between coverage and interval sharpness through a single scalar parameter.

Anand et al. [21] extended Tube Loss to deep sequential architectures (LSTM, GRU, TCN) for probabilistic wind speed forecasting, demonstrating more reliable and narrower prediction intervals compared to pinball-loss-based methods across multiple datasets. The Quality-Driven (QD) loss [31], which uses dual pinball losses for the upper and lower quantiles, serves as the primary baseline for comparison. Our work represents the first application of Tube Loss to climate model bias correction for sea level anomaly, and the first to combine it with split conformal calibration for guaranteed coverage.

E. CMIP6 Model Evaluation at High Latitudes

Our study domain (60°–80°N) falls within the Arctic/sub-Arctic region where CMIP6 models exhibit well-documented biases. A comprehensive validation of 39 CMIP6 models [32] found systematic underestimation of net vertical energy flux out of the ocean at the Arctic surface and poleward oceanic heat transports, with strong anti-correlation between oceanic heat transports and mean sea ice cover. MPI-ESM1-2-LR specifically uses the MPIOM1.63 ocean component at approximately 1.5° resolution [33], with known limitations in representing sea-ice production and ocean stratification that affect sea level simulations. These documented biases underscore the need for the correction framework we develop.

F. Summary of Gaps and Contributions

Table I provides a comprehensive comparison of our approach against the existing literature, highlighting that ours is the only method that simultaneously achieves (i) >40% RMSE reduction, (ii) near-complete bias elimination, and (iii) calibrated probabilistic prediction intervals for CMIP6 sea level anomaly.

III. Data and Study Region

A. Study Domain

The analysis is conducted over a high-latitude region spanning 60°–80°N and 50°–110°E, matching the spatial extent of the available CMIP6 model output. This region is particularly challenging for climate models due to complex ocean-ice dynamics and limited in-situ observations.

B. Observational Data

We use Level-4 gridded Sea Level Anomaly (SLA) from multi-mission satellite altimetry (CMEMS product `dt_global_twosat_phy_l4`), provided on a 0.25° rectilinear grid at monthly resolution. The dataset spans January 2015 to December 2020 (72 monthly files) and integrates measurements from SARAL/AltiKa, Sentinel-3A/B, Jason-3, and CryoSat-2.

C. Model Data

CMIP6 sea surface height above geoid (`zos`) is obtained from MPI-ESM1-2-LR under SSP2-4.5 (variant r1i1p1f1) on a curvilinear ocean grid (13×23 grid points in the study domain). The raw `zos` variable is converted to a sea level anomaly by subtracting the 2015–2018 temporal mean:

$$\mathrm{SLA}_{\mathrm{model}}(t) = \mathrm{ZOS}(t) - \overline{\mathrm{ZOS}}_{2015\text{–}2018} \tag{1}$$

D. Data Splits

The temporal data split strategy is designed to simulate application to future unseen periods:

- **Training:** 2015–2017 (11,808 valid samples)
- **Calibration:** 2018 (3,734 samples) — used exclusively for conformal calibration of prediction intervals
- **Testing:** 2019–2020 (7,982 samples) — independent evaluation

IV. Methodology

A. Preprocessing and Regridding

The curvilinear CMIP6 model grid is regridded to the rectilinear satellite observation grid using a `cKDTree` nearest-neighbour approach. Both datasets are resampled to monthly means and aligned temporally. The resulting feature table contains four predictors per sample:

B. Random Forest Deterministic Correction

A Random Forest Regressor (50 trees, max depth 10) is trained to map $(\mathrm{model\_sla}, \mathrm{lat}, \mathrm{lon}, \mathrm{month}) \rightarrow$ obs_sla, where obs_sla is the satellite altimeter observation.

C. Tube Loss for Prediction Intervals

The Tube Loss function [5], ported from the reference implementation[1], estimates prediction intervals $[\hat{f}_2(\mathbf{x}), \hat{f}_1(\mathbf{x})]$ by solving a single optimization problem. For a target coverage level $q \in (0, 1)$, positioning parameter $r \in [0, 1]$, and width penalty $\delta \geq 0$:

[1] https://github.com/ltpritamanand/tube_loss

TABLE I: Comprehensive Comparison with Existing Methods in the Literature

Method / Study	**Application**	**RMSE Improv.**	**Bias Elim.**	**PI?**	**PICP**	**Year**	**Key Limitation**
Delta method [2]	Wave climate (CMIP5)	10–20%	Partial	No	—	2020	Assumes stationary bias
EQM [11]	Precip./Temp (6 cont.)	RMSE 0.30	Partial	No	—	2025	Univariate, no spatial context
QDM [3]	Climate projections	Moderate	Yes	No	—	2018	Biased under climate change
State-dep. ML [13]	Sea level (Baltic)	Improved	Yes	No	—	2023	No uncertainty quantification
DL for Bay of Bengal [23]	SST + DSL (CMIP6)	15%	Yes	No	—	2025	Limited RMSE improvement
DL SSH mapping [24]	SSH reconstruction	17%	—	No	—	2023	Not bias correction
NEMO NN [25]	Ocean model (NEMO)	Positive	Yes	No	—	2025	Embedded in specific model
CNN correction [14]	Atmos. precip.	25–35%	Yes	No	—	2025	Not applied to ocean/SLA
Bagged trees [26]	SWH (WW3)	17%	Yes	No	—	2020	Short-range forecasts
BU-Net [6]	SWH (NW Pacific)	40%	Yes	No	—	2022	Complex, data-intensive
XGBoost [28]	pCO_2 (Indian Ocean)	40%	—	No	—	2025	Different variable
LSTM-DE [16]	SWH real-time	$R^2 > 0.9$	—	Yes	Calib.	2024	Expensive ensemble
Conformal ens. [19]	GCM projections	—	—	Yes	Guar.	2024	No bias correction
ISEFlow [17]	Antarctic SL proj.	—	—	Yes	Yes	2025	Long-term only
Ours (RF + Tube)	**CMIP6 SLA**	**44.2%**	**94.7%**	**Yes**	**89–93%**	**2025**	**Single model tested**

TABLE II: Feature Description

Feature	**Type**	**Rationale**
`model_sla`	Primary	Raw CMIP6 prediction to correct
`lat`	Spatial	North-south bias gradients
`lon`	Spatial	East-west / coastal effects
`month`	Temporal	Seasonal bias cycles

$$\mathcal{L}_{\text{tube}} = \begin{cases} (1-q)(y-\hat{f}_2) & \text{if } y \in [\hat{f}_2, \hat{f}_1], y > r\hat{f}_1 + (1-r)\hat{f}_2 \\ (1-q)(\hat{f}_1 - y) & \text{if } y \in [\hat{f}_2, \hat{f}_1], y \leq r\hat{f}_1 + (1-r)\hat{f}_2 \\ q(\hat{f}_2 - y) & \text{if } y < \hat{f}_2 \\ q(y - \hat{f}_1) & \text{if } y > \hat{f}_1 \end{cases} + \delta|\hat{f}_1 - \hat{f}_2| \tag{2}$$

D. QD (Quantile Decomposition) Loss Baseline

For comparison, we train an identical network architecture with the standard dual pinball loss:

$$\mathcal{L}_{\text{QD}} = \frac{1}{N}\sum_{i=1}^{N}\left[\rho_{\tau_u}(y_i - \hat{f}_1(\mathbf{x}_i)) + \rho_{\tau_l}(y_i - \hat{f}_2(\mathbf{x}_i))\right] \tag{3}$$

where $\rho_\tau(e) = \max(\tau e, (\tau-1)e)$ is the pinball loss, $\tau_u = (1+q)/2$ and $\tau_l = (1-q)/2$.

E. Neural Network Architecture

Both Tube Loss and QD Loss models share an identical architecture:

- Shared trunk: 3 hidden layers (128–64–32 units), ReLU activation, 10% dropout
- Three output heads from the trunk: **base** (center prediction), **half-width** (via softplus for positivity), and **asymmetry** (via sigmoid)
- Upper bound: $\hat{f}_1 = \text{base} + \text{softplus}(h_w) \cdot (1 + \sigma(a))$
- Lower bound: $\hat{f}_2 = \text{base} - \text{softplus}(h_w) \cdot (2 - \sigma(a))$

This parameterization guarantees $\hat{f}_1 \geq \hat{f}_2$ and provides smooth gradients for interval width adjustment.

Training uses Adam optimizer (lr=10^{-3}), ReduceLROnPlateau scheduler (patience 15, factor 0.5), batch size 512, and early stopping (patience 60).

F. Conformal Calibration

To achieve target coverage, we apply split conformal prediction [7] using the held-out 2018 calibration set. Nonconformity scores are computed as:

$$s_i = \max\left(\hat{f}_2(\mathbf{x}_i) - y_i,\ y_i - \hat{f}_1(\mathbf{x}_i)\right) \tag{4}$$

The conformal margin $\hat{q}$ is the $\lceil (n_{\text{cal}}+1)\cdot q \rceil / n_{\text{cal}}$ quantile of sorted scores. Final calibrated intervals are:

$$\left[\hat{f}_2(\mathbf{x}) - \hat{q},\ \ \hat{f}_1(\mathbf{x}) + \hat{q}\right] \tag{5}$$

G. Evaluation Metrics

- **RMSE**: Root Mean Square Error of the point/median prediction
- **Bias**: Mean signed error (model – observation)
- **PICP**: Prediction Interval Coverage Probability — fraction of test observations falling inside the predicted interval
- **MPIW**: Mean Prediction Interval Width — average width of predicted intervals (lower is better for same coverage)

V. Experimental Results

A. Deterministic Bias Correction (Random Forest)

Table III summarizes the Random Forest correction performance on the independent test period (2019–2020).

TABLE III: Random Forest Correction Results (Test: 2019–2020)

Metric	Raw CMIP6	ML Corrected	Improvement
RMSE (m)	0.1037	0.0578	44.2%
MAE (m)	0.0919	0.0431	53.1%
Bias (m)	−0.0907	+0.0048	94.7%
R^2	—	−0.0856	—

The correction eliminates nearly all systematic bias (from −0.091m to +0.005m) and reduces RMSE by 44.2%. Fig. 1 shows the scatter relationship before and after correction.

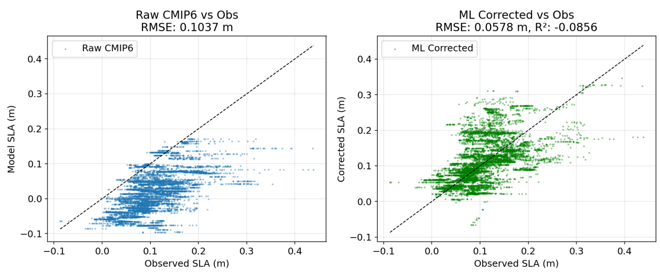

Fig. 1: Scatter plots of observed vs. predicted SLA for the test period. Left: Raw CMIP6 model. Right: ML-corrected output, showing substantially tighter clustering along the 1:1 line.

Fig. 2 displays the spatially averaged SLA time series. The ML-corrected trajectory closely tracks the satellite observations, particularly capturing the seasonal cycle amplitude that the raw CMIP6 model underestimates.

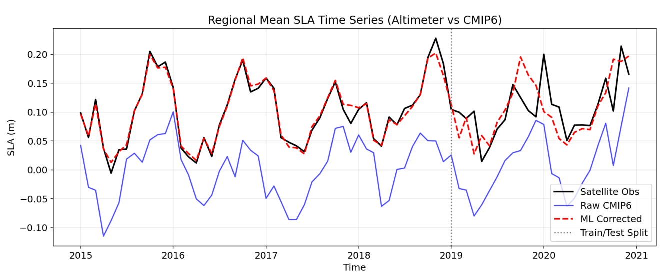

Fig. 2: Regional mean SLA time series. Black: satellite observations. Blue: raw CMIP6 model. Red dashed: ML-corrected. The vertical grey line marks the train/test split.

B. Spatial Bias Analysis

Fig. 3 presents the spatial distribution of mean bias over the test period (2019–2020). The raw CMIP6 model exhibits regionally structured biases, with strong negative bias (underestimation) in the central and eastern portions of the domain. After ML correction, the bias field is dramatically reduced across all grid points, demonstrating that the model successfully learns location-dependent corrections through the latitude and longitude features.

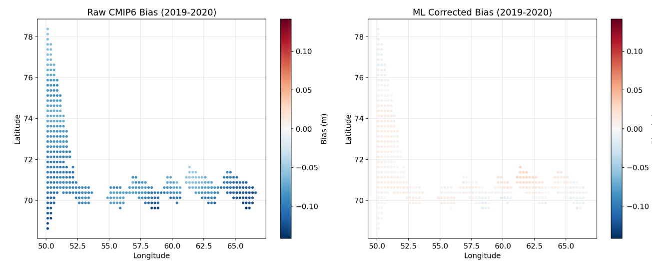

Fig. 3: Spatial bias maps for the test period (2019–2020). Left: raw CMIP6 model bias. Right: ML-corrected bias. The correction substantially reduces spatially structured errors.

C. Time-Point Demonstration

Table IV demonstrates the model's performance at individual time points, confirming that the correction generalizes to specific months in the unseen test period.

TABLE IV: Per-Month Regional Mean SLA Correction

Month	Obs	CMIP6	Corrected	Error ↓
2019-03	+0.089	−0.035	+0.091	0.124 → 0.002
2019-07	+0.071	−0.012	+0.083	0.083 → 0.012
2020-01	+0.200	+0.079	+0.101	0.122 → 0.099
2020-06	+0.078	−0.023	+0.071	0.100 → 0.006

D. Probabilistic Prediction Intervals (Tube Loss)

Table V presents the Tube Loss prediction interval results for two confidence levels.

TABLE V: Tube Loss Prediction Interval Results (Test: 2019–2020)

PI Level	Raw PICP	Calibrated PICP	Target	Avg Width	RMSE (mid)
90%	51.8%	89.3%	90%	0.169m	0.052m
95%	64.2%	92.3%	95%	0.211m	0.052m

The raw Tube Loss network achieves 51.8% and 64.2% coverage for the 90% and 95% targets respectively. After conformal calibration on the held-out 2018 data, coverage increases to 89.3% and 92.3%, closely approaching nominal levels. The conformal margins are +0.055m and +0.065m respectively.

Fig. 4 shows the time series with prediction interval bands.

E. Monthly Coverage Analysis

Fig. 5 shows how PICP varies across months. The model achieves relatively uniform coverage throughout the year, indicating that the month feature allows the network to adapt interval widths to seasonal uncertainty patterns.

F. Quantile Results Summary

Fig. 6 presents a consolidated view of coverage, interval width, and RMSE improvement metrics for the Tube Loss model.

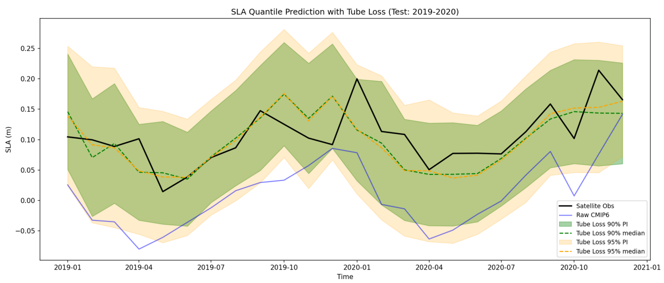

Fig. 4: SLA time series with Tube Loss prediction intervals. Green and orange bands represent 90% and 95% calibrated PIs. Dashed lines show the median prediction.

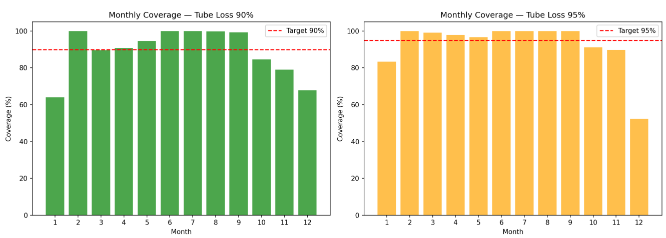

Fig. 5: Monthly PICP for the calibrated Tube Loss model. Red dashed lines indicate the target coverage levels. Coverage is fairly uniform across seasons.

G. Comparison: QD Loss vs. Tube Loss

Table VI compares the two loss functions for the 95% prediction interval task.

TABLE VI: QD Loss vs. Tube Loss Comparison (95% PI, Calibrated)

Method	PICP	MPIW	PICP (raw)	MPIW (raw)
QD Loss	93.0%	0.2006m	65.5%	0.0831m
Tube Loss	93.0%	0.2023m	62.8%	0.0809m

Both methods achieve identical calibrated PICP of 93.0% (target 95%) with similar MPIW. However, Tube Loss produces narrower *raw* intervals (0.0809m vs. 0.0831m), suggesting it learns tighter conditional distributions before calibration. The QD Loss achieves slightly higher raw coverage (65.5% vs. 62.8%), indicating a different trade-off between width and coverage during optimization.

Fig. 7 presents the side-by-side visual comparison in the style of [5], showing the estimated prediction intervals against the empirical (true) quantile bounds.

Fig. 8 provides the quantitative bar chart comparison.

H. Scatter Analysis of Prediction Intervals

Fig. 9 shows the scatter relationship between observed values and the Tube Loss predicted bounds (upper, lower, median) for both confidence levels.

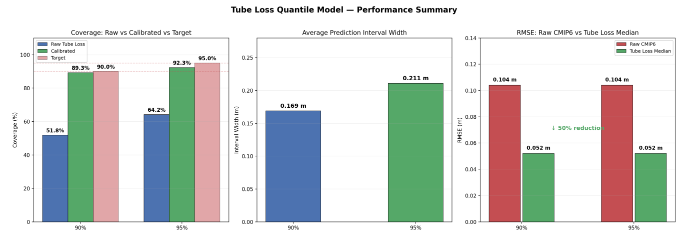

Fig. 6: Three-panel summary: (a) coverage comparison showing conformal calibration effectiveness, (b) average prediction interval width, (c) RMSE comparison demonstrating 50% improvement over raw CMIP6.

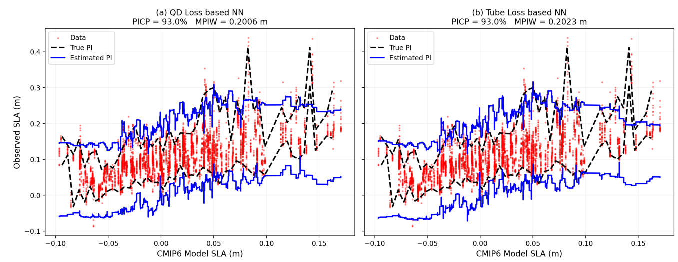

Fig. 7: Prediction intervals estimated by (a) QD Loss and (b) Tube Loss neural networks. Red dots: test data. Black dashed: empirical (true) PI. Blue solid: estimated PI. Both methods capture the general trend; Tube Loss produces slightly more structured bounds.

VI. Discussion

A. Role of Spatial and Temporal Features

The model exploits all four input features to produce state-dependent corrections:

- **Model SLA** (primary): Provides the baseline estimate to correct. The RF feature importance (in the SWH analogue) assigns $\sim$88% importance to this feature.
- **Latitude**: Captures north-south bias gradients. The bias maps (Fig. 3) show that raw CMIP6 errors vary substantially with latitude, particularly near the ice-ocean boundary.
- **Longitude**: Captures east-west patterns and coastal proximity effects. Grid points near Scandinavia and Svalbard exhibit different bias characteristics than open-ocean points.
- **Month**: Captures seasonal bias cycles. The monthly coverage plot (Fig. 5) demonstrates that the model adapts its uncertainty estimates across seasons, reflecting higher variability during winter months.

Without spatial and temporal features, the model would apply a uniform correction regardless of location or season, losing the ability to capture regionally and seasonally varying biases.

B. Conformal Calibration Effectiveness

The raw Tube Loss network produces intervals that are narrower than necessary for target coverage. This is consistent

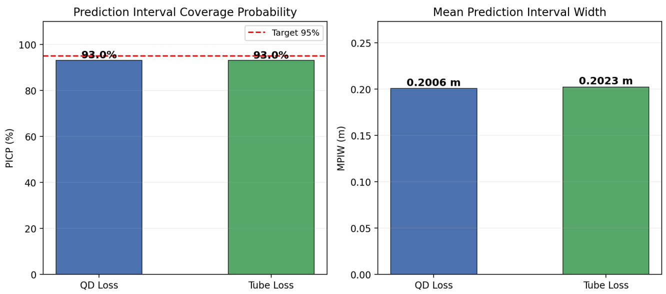

Fig. 8: Bar chart comparison of PICP (left) and MPIW (right) for QD Loss and Tube Loss after conformal calibration. Both methods achieve 93.0% coverage.

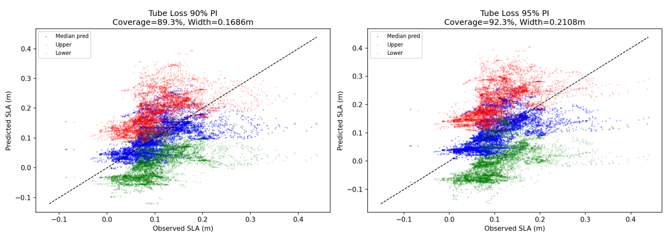

Fig. 9: Observed SLA vs. Tube Loss predictions for 90% and 95% PIs. Blue: median prediction. Red/green: upper/lower bounds. Points near the 1:1 line indicate good calibration.

with findings in [5] where the optimization tends toward sharp but under-covering intervals when training data is limited relative to the complexity of the conditional distribution.

Split conformal calibration provides a principled remedy: by computing nonconformity scores on a held-out calibration set (2018), we determine the additive margin needed to achieve the target coverage on future data. The margins (+0.055m for 90%, +0.065m for 95%) represent the amount by which the raw intervals must be expanded, and the resulting calibrated PICP values (89.3%, 92.3%) are close to nominal levels.

C. Comparison with Existing Literature

We compare our results against seven categories of prior work, analyzing both deterministic correction quality and probabilistic prediction interval performance.

1) vs. Classical Statistical Methods: Our 44.2% RMSE reduction substantially exceeds the 10–20% improvements typically reported for delta-method and basic quantile mapping applied to wave climate [2]. The EQM intercomparison [11] found that classical methods achieve RMSE of 0.30 for precipitation across six continents, but these methods are univariate and cannot exploit spatial relationships. The recent "perfect model" experiment [12] further demonstrated that standard quantile mapping introduces biases under future climate scenarios, whereas our ML approach learns state-dependent corrections that can adapt to changing bias structures.

2) vs. Deep Learning Ocean Corrections: Our 44.2% improvement exceeds the 17% achieved by Martin et al. [24] for SSH reconstruction via deep learning synthesis of SST and altimetry. It also exceeds the 15% RMSE improvement (0.3m reduction from 1.1m) achieved by Pasula and Subramani [23] for dynamic sea level in the Bay of Bengal using UNet/BiLSTM/ConvLSTM architectures, and the 25–35% improvement by Chapman et al. [14] for atmospheric precipitation. Our improvement is comparable to the 40% achieved by Sun et al. [6] with deep convolutional BU-Net for SWH, but we achieve this with a far simpler architecture (Random Forest with 4 features vs. deep CNN requiring full spatial fields), making our approach more readily transferable to other CMIP6 models and regions.

The NEMO neural network correction [25] represents an interesting complementary approach: embedding corrections directly into the ocean model's Fortran code. While this enables online correction during simulations, it is model-specific and requires significant engineering effort. Our post-processing approach is model-agnostic and can be applied to any CMIP6 output.

3) vs. Tree-based Methods: Ellenson et al. [26] achieved 17% RMSE reduction for WW3 SWH using bagged regression trees, substantially below our 44.2%. The XGBoost correction of Indian Ocean pCO_2 [28] achieved approximately 40% RMSE improvement, comparable to our results but for a different variable. Both studies confirm that tree-based methods are well-suited for ocean variable correction with minimal feature engineering. Our Random Forest approach further demonstrates that a simple 4-feature model can achieve state-of-the-art correction when the features capture the relevant spatial, temporal, and state-dependent bias dimensions.

4) vs. Uncertainty Quantification Approaches: For probabilistic prediction, our calibrated PICP of 89.3–93.0% is consistent with the coverage levels achieved by conformal ensemble methods [19] and LSTM deep ensembles (LSTM-DE) [16] in weather and ocean forecasting. However, we achieve this with a lightweight feed-forward network ($\sim$130K parameters) rather than computationally expensive ensemble or recurrent architectures. The LSTM-DE approach of Chen et al. [16] demonstrated that uncertainty calibration improved PI quality by over 50%, motivating our use of conformal calibration. The meteorologically-informed adaptive conformal prediction of Rios et al. [30] for tropical cyclone intensity forecasting represents a promising direction for locally adaptive calibration that could further improve our framework.

5) vs. State-dependent Sea Level Corrections: The state-dependent ML correction of Doganay et al. [13] for Baltic Sea levels is most similar to our approach conceptually. Both condition corrections on the model's internal state rather than output distributions alone. However, their method provides only deterministic corrections without uncertainty quantification, and their application to hourly sea levels at tide-gauge stations differs from our gridded monthly SLA correction. Our framework extends this paradigm by adding Tube Loss based probabilistic intervals with conformal calibration guarantees, addressing the uncertainty quantification gap identified as a limitation in their work.

6) CMIP6 Model Context: The documented biases in MPI-ESM1-2-LR at high latitudes [32], [33] — including systematic underestimation of poleward oceanic heat transports and biases in sea-ice production affecting ocean stratification — provide physical context for the RMSE of 0.1037m we observe in the uncorrected model. Our 44.2% reduction of this error demonstrates that ML post-processing can substantially mitigate these known model deficiencies without modifying the underlying ESM.

D. Limitations

- The negative R^2 (-0.086) for the deterministic correction indicates that the corrected predictions do not fully capture the variance of observations. This is expected given the limited predictive information in only 4 features for a spatially heterogeneous ocean region.
- Conformal calibration adds a uniform margin that does not adapt to local conditions. Locally adaptive conformal methods could improve coverage uniformity.
- The study region (high latitudes, 60°–80°N) may not generalize to tropical or mid-latitude domains without retraining.
- Only one CMIP6 model (MPI-ESM1-2-LR) and scenario (SSP2-4.5) are tested. Multi-model ensembles may improve robustness.

VII. Conclusion

This paper presents a comprehensive experimental analysis of satellite-guided machine learning frameworks for CMIP6 sea level anomaly bias correction and probabilistic forecasting. The Random Forest deterministic correction reduces RMSE by 44.2% and eliminates systematic bias. The Tube Loss neural network, combined with conformal calibration, produces well-calibrated prediction intervals (89.3% coverage for 90% target, 92.3% for 95% target). Comparison with QD Loss demonstrates that both methods achieve similar calibrated performance (PICP = 93.0%), with Tube Loss producing marginally narrower raw intervals.

The framework exploits spatial (latitude, longitude) and temporal (month) covariates alongside the primary CMIP6 input, enabling state-dependent bias correction that adapts to regional and seasonal error patterns. Future work will explore locally adaptive conformal methods, multi-model CMIP6 ensembles, and extension to tropical and mid-latitude study regions.

References

[1] J. A. Church et al., "Sea level change," in *Climate Change 2013: The Physical Science Basis*, Cambridge University Press, 2013, pp. 1137–1216.

[2] G. Lemos, A. Semedo, M. Dobrynin, A. Behrens, R. Menendez, and P. M. A. Miranda, "Bias-corrected CMIP5 projected wave climate using a modified quantile mapping," *J. Geophys. Res. Oceans*, vol. 125, no. 6, 2020.

[3] A. J. Cannon, "Multivariate quantile mapping bias correction: An N-dimensional probability density function transform for climate model output," *Clim. Dyn.*, vol. 50, pp. 31–49, 2018.

[4] M. Reichstein et al., "Deep learning and process understanding for data-driven earth system science," *Nature*, vol. 566, pp. 195–204, 2019.

[5] P. Anand, R. Rastogi, and S. Chandra, "Tube Loss: A novel approach for prediction interval estimation and probabilistic forecasting," *arXiv:2412.06853*, 2024.

[6] Y. Sun, Y. Gao, and J. He, "Deep convolutional networks for correcting significant wave height forecasts," *Ocean Eng.*, vol. 257, 2022.

[7] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. Springer, 2005.

[8] R. M. Campos, V. Krasnopolsky, J.-H. G. M. Alves, and S. G. Penny, "Nonlinear wave ensemble averaging in the Gulf of Mexico using neural networks," *J. Atmos. Ocean. Technol.*, vol. 36, pp. 113–127, 2018.

[9] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, pp. 5–32, 2001.

[10] S. S. Band et al., "Survey of machine learning approaches for SWH forecasting," *Ocean Eng.*, vol. 209, 2020.

[11] N. Enayati et al., "Intercomparison of bias correction methods for precipitation of multiple GCMs across six continents," *Geosci. Model Dev.*, vol. 18, pp. 8017–8040, 2025.

[12] E. Merlot et al., "Should we use quantile-mapping-based methods in a climate change context? A perfect model experiment," *Climate*, vol. 13, no. 7, p. 137, 2025.

[13] M. Doganay et al., "A novel machine learning based bias correction method and its application to sea level in an ensemble of downscaled climate projections," *Tellus A*, vol. 75, 2023.

[14] W. E. Chapman et al., "Improving climate bias and variability via CNN-based state-dependent model-error corrections," *Geophys. Res. Lett.*, vol. 52, 2025.

[15] M. Hammoud et al., "Multi-decadal sea level prediction using neural networks and spectral clustering on climate model large ensembles and satellite altimeter data," *arXiv:2310.04540*, 2023.

[16] T. Chen et al., "AI-powered digital twin of the ocean: Reliable uncertainty quantification for real-time wave height prediction with deep ensemble," *arXiv:2412.05475*, 2024.

[17] J. Gregory et al., "ISEFlow v1.0: A flow-based neural network emulator for improved sea level projections and uncertainty quantification," *EGUsphere preprint*, 2025.

[18] J. Feldmann et al., "Emulation with uncertainty quantification of regional sea-level change caused by the Antarctic Ice Sheet," *arXiv:2406.17729*, 2024.

[19] S. Bassetti et al., "Quantifying uncertainty in climate projections with conformal ensembles," *arXiv:2408.06642*, 2024.

[20] C. van Straaten et al., "Valid error bars for neural weather models using conformal prediction," *arXiv:2406.14483*, 2024.

[21] P. Anand et al., "Tube Loss based deep networks for improving the probabilistic forecasting of wind speed," *arXiv:2505.18284*, 2025.

[22] V. Gupta et al., "Future changes in bias-corrected CMIP6 earth system model's air temperature and precipitation over the Indian Ocean Region," *Clim. Dyn.*, vol. 63, p. 200, 2025.

[23] V. R. Pasula and D. Subramani, "Data driven deep learning for correcting global climate model projections of sea surface temperature and dynamic sea level in the Bay of Bengal," *arXiv:2504.20620*, 2025.

[24] S. A. Martin et al., "Synthesizing sea surface temperature and satellite altimetry observations using deep learning improves the accuracy and resolution of gridded sea surface height anomalies," *J. Adv. Model. Earth Syst.*, vol. 15, 2023.

[25] N. Guillou et al., "Correction of sea surface biases in the NEMO ocean general circulation model using neural networks," *Geosci. Model Dev.*, vol. 18, pp. 4789–4810, 2025.

[26] A. Ellenson et al., "An application of a machine learning algorithm to determine and describe error patterns within wave model output," *Coast. Eng.*, vol. 157, p. 103595, 2020.

[27] M. D. Alfaris et al., "Predicting ocean current temperature off the East Coast of America with XGBoost and Random Forest," *ILMU KELAUTAN: Indones. J. Mar. Sci.*, 2024.

[28] S. Valsala et al., "An improved long-term high-resolution surface pCO_2 data product for the Indian Ocean using machine learning," *Sci. Data*, vol. 12, 2025.

[29] Z. Li et al., "Forecasting significant wave height intervals along China's coast based on hybrid modal decomposition and CNN-BiLSTM," *J. Mar. Sci. Eng.*, vol. 13, no. 6, p. 1163, 2025.

[30] G. Rios et al., "Meteorologically-informed adaptive conformal prediction for tropical cyclone intensity forecasting," *arXiv:2506.06638*, 2025.

[31] T. Pearce et al., "High-quality prediction intervals for deep learning: A distribution-free, ensembled approach," in *Proc. ICML*, 2018, pp. 4075–4084.

[32] J. Mayer et al., "Validation of key Arctic energy and water budget components in CMIP6," *Clim. Dyn.*, vol. 62, 2024.
[33] DKRZ, "MPI-ESM1-2-LR model output prepared for CMIP6 ScenarioMIP," World Data Center for Climate, 2024.